

Robust last-success stopping with predicted odds

Detailed notes: independent discrete trials and the continuous Poisson limit

Abstract

We study the problem of stopping on the last success. In the independent finite model, Bruss' odds theorem says that the optimal policy is obtained by summing the true odds from the end until reaching one. We then assume that only predicted odds are available and that the predictions are multiplicatively ρ -accurate. We prove finite- n robustness bounds for threshold rules that sum predicted odds up to a target θ , optimize this target, and compare it with the naive target $\theta = 1$. We also clarify that this optimized threshold is not a global minimax rule over randomized strategies, give an exact two-period counterexample, and record the corresponding semi-infinite linear programming formulation. We then derive the analogous and cleaner result for the continuous Poisson model, where the state is summarized by the remaining intensity mass. Finally, we add a Bayesian model in which local multiplicative errors are drawn independently from a known law; in that case the optimal rule is again exactly Bruss' rule, applied to induced effective odds.

1 The basic last-success problem

Let $I_1, \dots, I_n$ be Bernoulli indicators. A success occurs at time i when $I_i = 1$. The goal is to choose a stopping time τ and maximize

$$\mathbb{P}(I_\tau = 1, I_{\tau+1} = \dots = I_n = 0),$$

that is, the probability of stopping exactly on the last success.

In the independent model, write

$$p_i = \mathbb{P}(I_i = 1), \quad q_i = 1 - p_i, \quad r_i = \frac{p_i}{q_i}.$$

The number r_i is the *odd* of a success at time i . For $k \in \{1, \dots, n\}$, define the true tail odds and the true tail failure product by

$$R_k = \sum_{j=k}^n r_j, \quad Q_k = \prod_{j=k}^n q_j = \prod_{j=k}^n \frac{1}{1 + r_j}.$$

We also use the convention $R_{n+1} = 0$ and $Q_{n+1} = 1$.

Lemma 1 (Value of a deterministic threshold). *Assume that $I_1, \dots, I_n$ are independent. If the policy ignores all successes before k and then stops at the first success in $\{k, k+1, \dots, n\}$, its success probability is*

$$W(k) = Q_k R_k.$$

Proof. The policy wins if and only if there is exactly one success in the tail $\{k, \dots, n\}$. By

independence,

$$\begin{aligned}\mathbb{P}(\text{exactly one success in } \{k, \dots, n\}) &= \sum_{i=k}^n p_i \prod_{\substack{j=k \\ j \neq i}}^n q_j \\ &= \left(\prod_{j=k}^n q_j \right) \sum_{i=k}^n \frac{p_i}{q_i} \\ &= Q_k R_k.\end{aligned}$$

This proves the claim. $\square$

Theorem 1 (Bruss' odds theorem, threshold form). *Assume that the indicators are independent. In the non-degenerate case $R_1 > 1$, an optimal rule is the following: let*

$$s = \sup \left\{ k \in \{1, \dots, n\} : R_k = \sum_{j=k}^n r_j \geq 1 \right\}.$$

Equivalently,

$$R_s \geq 1 > R_{s+1}.$$

Then it is optimal to ignore all successes before s , and to stop at the first success from s onward. The optimal value is

$$V^* = Q_s R_s.$$

If $R_1 < 1$, then the optimal rule starts at 1, i.e. it stops at the first success.

The theorem is due to Bruss (2000). The main message is that one sums *odds*, not success probabilities. When all individual probabilities are small, odds and probabilities are close; this is why the Poisson limit below has a particularly simple form.

2 Predicted odds in the discrete independent model

We now suppose that the true odds r_i are not known exactly. Instead we have predicted odds $\bar{r}_i$, and we know that the prediction error is multiplicative:

$$\boxed{\frac{1}{\rho} \bar{r}_i \leq r_i \leq \rho \bar{r}_i} \quad \text{for every } i, \tag{A}$$

where $\rho \geq 1$ is known. Equivalently,

$$\max \left\{ \frac{r_i}{\bar{r}_i}, \frac{\bar{r}_i}{r_i} \right\} \leq \rho.$$

Let

$$\bar{R}_k = \sum_{j=k}^n \bar{r}_j.$$

Definition 1 (Predicted-threshold rule). *Fix a target $\theta > 0$. The rule Alg_θ selects*

$$\hat{s}_\theta = \begin{cases} \sup\{k : \bar{R}_k \geq \theta\}, & \text{if such a } k \text{ exists,} \\ 1, & \text{otherwise.} \end{cases}$$

It then ignores all successes before $\hat{s}_\theta$ and stops at the first success from $\hat{s}_\theta$ onward.

Thus $\theta = 1$ is the naive predicted-Bruss rule: it applies Bruss' rule to the predicted odds as if the prediction were exact. The robust question is whether another target θ , depending on ρ , gives a better worst-case guarantee.

Throughout this section, we assume the true instance is non-degenerate:

$$R_1 > 1.$$

Let s be the true Bruss threshold and let $k = \hat{s}_\theta$ be the threshold chosen from predictions. The true optimal value is

$$V^\star = Q_s R_s,$$

whereas the true success probability of Alg_θ is

$$W_\theta = Q_k R_k.$$

2.1 Basic consequences of multiplicative prediction error

Lemma 2 (Tail constraints induced by prediction accuracy). *Let $x = \rho\theta$. Suppose that $1 \leq x \leq \rho^2$. Then the threshold $k = \hat{s}_\theta$ selected by Alg_θ satisfies*

$$R_k \geq \frac{\theta}{\rho} = \frac{x}{\rho^2}, \quad R_{k+1} < \rho\theta = x.$$

Proof. There are two cases.

First suppose that the predicted target is reached, i.e. that there exists some k with $\bar{R}_k \geq \theta$, and let $k = \hat{s}_\theta$. By definition of k ,

$$\bar{R}_k \geq \theta \quad \text{and} \quad \bar{R}_{k+1} < \theta,$$

where $\bar{R}_{n+1} = 0$. Summing the pointwise inequalities $r_j \geq \bar{r}_j/\rho$ and $r_j \leq \rho\bar{r}_j$ over tails gives

$$R_k \geq \frac{1}{\rho} \bar{R}_k \geq \frac{\theta}{\rho},$$

and

$$R_{k+1} \leq \rho \bar{R}_{k+1} < \rho\theta.$$

Now suppose that the predicted target is never reached. Then $k = 1$. Since $R_1 > 1$ and $x \leq \rho^2$,

$$R_k = R_1 > 1 \geq \frac{x}{\rho^2} = \frac{\theta}{\rho}.$$

Moreover $\bar{R}_1 < \theta$, hence $\bar{R}_2 \leq \bar{R}_1 < \theta$, and therefore

$$R_{k+1} = R_2 \leq \rho \bar{R}_2 < \rho\theta = x.$$

This proves both inequalities in all cases. □

2.2 When the predicted rule starts too late

The event $k \geq s$ means that the predicted rule starts no earlier than the true optimal threshold. It may miss some useful successes, but it cannot be hurt by extra early false stops.

Lemma 3 (Late-start bound). *If $k \geq s$, then*

$$\frac{W_\theta}{V^\star} \geq \frac{x}{\rho^2}.$$

Proof. If $k = s$, the ratio is 1, and the claim follows since $x \leq \rho^2$. Assume now that $k > s$. Split the true optimal tail $\{s, \dots, n\}$ into the part discarded by Alg_θ , namely $\{s, \dots, k-1\}$, and the part kept by Alg_θ , namely $\{k, \dots, n\}$. Define

$$A = \sum_{j=s}^{k-1} r_j, \quad R = R_k, \quad P = \prod_{j=s}^{k-1} (1 + r_j).$$

Then $R_s = A + R$ and $Q_s = Q_k/P$, because

$$Q_s = \prod_{j=s}^{k-1} \frac{1}{1 + r_j} \prod_{j=k}^n \frac{1}{1 + r_j} = \frac{Q_k}{P}.$$

Therefore

$$V^\star = Q_s R_s = \frac{Q_k}{P} (A + R), \quad W_\theta = Q_k R.$$

Thus

$$\frac{W_\theta}{V^\star} = \frac{RP}{A + R}.$$

Since all $r_j \geq 0$, the product dominates one plus the sum:

$$P = \prod_{j=s}^{k-1} (1 + r_j) \geq 1 + \sum_{j=s}^{k-1} r_j = 1 + A.$$

Consequently

$$\frac{W_\theta}{V^\star} \geq \frac{R(1 + A)}{A + R}.$$

We now show that the last expression is at least $\min\{R, 1\}$. If $R \leq 1$, then

$$\frac{R(1 + A)}{A + R} \geq R \iff 1 + A \geq A + R \iff 1 \geq R,$$

which holds. If $R \geq 1$, then

$$\frac{R(1 + A)}{A + R} \geq 1 \iff R + RA \geq A + R \iff A(R - 1) \geq 0,$$

which also holds. Hence

$$\frac{W_\theta}{V^\star} \geq \min\{R, 1\}.$$

By the tail-constraints lemma,

$$R = R_k \geq \frac{x}{\rho^2}.$$

Since $x \leq \rho^2$, the number x/ρ^2 is at most 1. Therefore

$$\min\{R, 1\} \geq \frac{x}{\rho^2},$$

which proves the lemma. □

2.3 When the predicted rule starts too early

The event $k < s$ means that the predicted rule accepts too early. Then the rule is exposed to the risk of stopping on a success that is followed by another success. The next lemma bounds the worst possible multiplicative loss from the extra early region.

Lemma 4 (Early-start bound). *If $k < s$, then, for $x = \rho\theta \geq 1$,*

$$\frac{W_\theta}{V^\star} \geq \left(1 + \frac{x-1}{n-1}\right)^{-(n-1)}.$$

Proof. Write

$$a = r_k, \quad B = \sum_{j=k+1}^{s-1} r_j, \quad R = R_s,$$

and

$$P_B = \prod_{j=k+1}^{s-1} (1 + r_j),$$

with the usual convention that an empty product is 1. Then

$$R_k = a + B + R,$$

and

$$Q_k = \frac{Q_s}{(1+a)P_B}.$$

Hence

$$\frac{W_\theta}{V^\star} = \frac{Q_k R_k}{Q_s R_s} = \frac{a + B + R}{(1+a)P_B R}.$$

Since $R = R_s \geq 1$, we have $B + R \geq 1$, and therefore

$$a + B + R \geq 1 + a.$$

It follows that

$$\frac{W_\theta}{V^\star} \geq \frac{1}{P_B R}.$$

It remains to upper-bound $P_B R$. Because $k < s$, the index $k+1$ still lies before or at the true optimal tail. the tail-constraints lemma gives

$$R_{k+1} < x.$$

But

$$R_{k+1} = \sum_{j=k+1}^n r_j = B + R.$$

Thus

$$B + R < x.$$

Let

$$m = s - k - 1.$$

There are m factors in P_B and therefore $m+1$ nonnegative factors in $P_B R$, namely

$$R, \quad 1 + r_{k+1}, \dots, 1 + r_{s-1}.$$

Their sum is

$$R + \sum_{j=k+1}^{s-1} (1 + r_j) = R + m + B < m + x.$$

By the arithmetic–geometric mean inequality,

$$P_B R \leq \left(\frac{m+x}{m+1} \right)^{m+1} = \left(1 + \frac{x-1}{m+1} \right)^{m+1}.$$

Now $m+1 = s-k \leq n-1$, because $1 \leq k < s \leq n$. For $x \geq 1$, the function

$$y \mapsto \left(1 + \frac{x-1}{y} \right)^y, \quad y > 0,$$

is increasing. One way to see this is to differentiate its logarithm:

$$\frac{d}{dy} \left[y \log \left(1 + \frac{c}{y} \right) \right] = \log \left(1 + \frac{c}{y} \right) - \frac{c}{y+c} \geq 0, \quad c = x-1 \geq 0,$$

since $\log(1+u) \geq u/(1+u)$ for $u \geq 0$. Therefore

$$P_B R \leq \left(1 + \frac{x-1}{n-1} \right)^{n-1}.$$

Taking reciprocals gives

$$\frac{W_\theta}{V^\star} \geq \frac{1}{P_B R} \geq \left(1 + \frac{x-1}{n-1} \right)^{-(n-1)}.$$

□

2.4 Optimizing the robust target

Theorem 2 (Robust target rule for independent discrete trials). *Let $R_1 > 1$, and assume the odds prediction error (A). Fix $\theta > 0$ and set*

$$x = \rho\theta.$$

If $x \in [1, \rho^2]$, then the predicted-threshold rule Alg_θ satisfies

$$\boxed{\frac{W_\theta}{V^\star} \geq \min \left\{ \frac{x}{\rho^2}, \left(1 + \frac{x-1}{n-1} \right)^{-(n-1)} \right\}.} \quad (1)$$

The lower bound in (1) is maximized by the unique solution $x_n^\star(\rho) \in [1, \rho^2]$ of

$$\boxed{x \left(1 + \frac{x-1}{n-1} \right)^{n-1} = \rho^2.} \quad (2)$$

Equivalently, the optimized robust target is

$$\boxed{\theta_n^\star(\rho) = \frac{x_n^\star(\rho)}{\rho}.} \quad (3)$$

The corresponding guaranteed ratio is

$$\boxed{\alpha_n(\rho) = \frac{x_n^\star(\rho)}{\rho^2}.} \quad (4)$$

Proof. Let s be the true Bruss threshold and $k = \hat{s}_\theta$ the threshold selected from the predicted odds. Exactly one of the two cases holds.

If $k \geq s$, the late-start lemma gives

$$\frac{W_\theta}{V^\star} \geq \frac{x}{\rho^2}.$$

If $k < s$, the early-start lemma gives

$$\frac{W_\theta}{V^\star} \geq \left(1 + \frac{x-1}{n-1}\right)^{-(n-1)}.$$

Combining these two alternatives gives (1).

It remains to optimize the right-hand side of (1). The first term,

$$L(x) = \frac{x}{\rho^2},$$

is continuous and strictly increasing on $[1, \rho^2]$. The second term,

$$E(x) = \left(1 + \frac{x-1}{n-1}\right)^{-(n-1)},$$

is continuous and strictly decreasing on $[1, \rho^2]$. Therefore the maximum of $\min\{L(x), E(x)\}$ is attained either at their unique crossing point or at an endpoint. At $x = 1$,

$$L(1) = \frac{1}{\rho^2} \leq 1 = E(1),$$

whereas at $x = \rho^2$,

$$L(\rho^2) = 1 \geq E(\rho^2).$$

Thus a crossing exists. It is unique because L is strictly increasing and E is strictly decreasing. The equality $L(x) = E(x)$ is

$$\frac{x}{\rho^2} = \left(1 + \frac{x-1}{n-1}\right)^{-(n-1)},$$

which is equivalent to (2). At this crossing, the common value is x/ρ^2 , proving (3) and (4). $\square$

Remark 1 (What is optimized?). *The theorem optimizes the target θ inside the natural one-parameter family of Bruss-type rules that sum predicted odds until a fixed target. It does not assert that no more complicated randomized or history-dependent rule could improve the global minimax constant. The point is that within the interpretable threshold family, the best robust target is explicit.*

2.5 Threshold optimality versus randomized minimax rules

The optimized target $\theta_n^\star(\rho)$ in Theorem 2 should be read in a precise sense: it maximizes the worst-case *lower bound* obtained for the one-parameter family of deterministic predicted-odds threshold rules. It is not, in general, a global minimax rule over all randomized stopping rules. This distinction does not contradict Bruss' theorem: for a fixed known instance, randomization cannot beat the Bruss optimum, since a randomized rule is a mixture of deterministic rules. However, in the robust problem the objective is a worst-case ratio over an uncertainty set, and randomization can improve that worst-case ratio.

The following two-period example makes the point explicit.

Proposition 1 (A two-period randomized minimax example). *Let*

$$n = 2, \quad \bar{r}_1 = \bar{r}_2 = 1, \quad \rho = 2.$$

Thus the true odds satisfy

$$r_1, r_2 \in \left[\frac{1}{2}, 2 \right].$$

Any randomized stopping rule is described by a single parameter $a \in [0, 1]$: if $I_1 = 1$, stop with probability a ; if the rule has not stopped and $I_2 = 1$, stop at time 2. The worst-case ratio of this rule is

$$\Gamma(a) = \min \left\{ 1 - \frac{a}{3}, \frac{2a+3}{5} \right\}.$$

Consequently the randomized minimax rule is

$$a^* = \frac{6}{11},$$

and its worst-case ratio is

$$\Gamma(a^*) = \frac{9}{11}.$$

By contrast, the two deterministic threshold choices have ratios $3/5$ and $2/3$. Hence a randomized rule is strictly better in this robust minimax sense.

Proof. For $i = 1, 2$, write

$$p_i = \frac{r_i}{1+r_i}, \quad q_i = \frac{1}{1+r_i}.$$

A randomized rule for a two-period problem only has one nontrivial decision: when $I_1 = 1$, it stops with some probability $a \in [0, 1]$. If it reaches time 2, it is clearly optimal to stop whenever $I_2 = 1$. Its success probability is

$$W_a(r_1, r_2) = ap_1q_2 + q_1p_2 + (1-a)p_1p_2 = p_2 + ap_1(q_2 - p_2).$$

The clairvoyant Bruss value is

$$V^*(r_1, r_2) = \begin{cases} p_2, & r_2 \geq 1, \\ q_1q_2(r_1 + r_2), & r_2 < 1. \end{cases}$$

First suppose that $r_2 \geq 1$. Then

$$\frac{W_a}{V^*} = 1 - a \frac{r_1(r_2 - 1)}{(1+r_1)r_2}.$$

The subtracted factor is increasing in both r_1 and r_2 , so the worst case on $[1/2, 2]^2 \cap \{r_2 \geq 1\}$ is attained at $r_1 = r_2 = 2$. This gives

$$\frac{W_a}{V^*} \geq 1 - \frac{a}{3}.$$

Now suppose that $r_2 < 1$. Then

$$\frac{W_a}{V^*} = \frac{r_2 + r_1r_2 + ar_1(1-r_2)}{r_1 + r_2}.$$

For fixed $r_2 < 1$, this expression is nonincreasing in r_1 , hence its minimum is attained at $r_1 = 2$. With $r_1 = 2$, it becomes

$$\frac{2a + r_2(3 - 2a)}{2 + r_2},$$

which is nondecreasing in r_2 . Hence the worst case in the region $r_2 < 1$ is attained at $r_2 = 1/2$, and gives

$$\frac{W_a}{V^*} \geq \frac{2a+3}{5}.$$

Both lower bounds are tight, being attained at admissible endpoints. Therefore

$$\Gamma(a) = \min \left\{ 1 - \frac{a}{3}, \frac{2a+3}{5} \right\}.$$

The first term is decreasing in a , the second is increasing in a , so the maximum of their minimum is obtained by equalizing them:

$$1 - \frac{a}{3} = \frac{2a+3}{5}.$$

This yields $a^* = 6/11$, and the common value is $9/11$. The deterministic choices $a = 0$ and $a = 1$ give ratios $3/5$ and $2/3$, respectively. $\square$

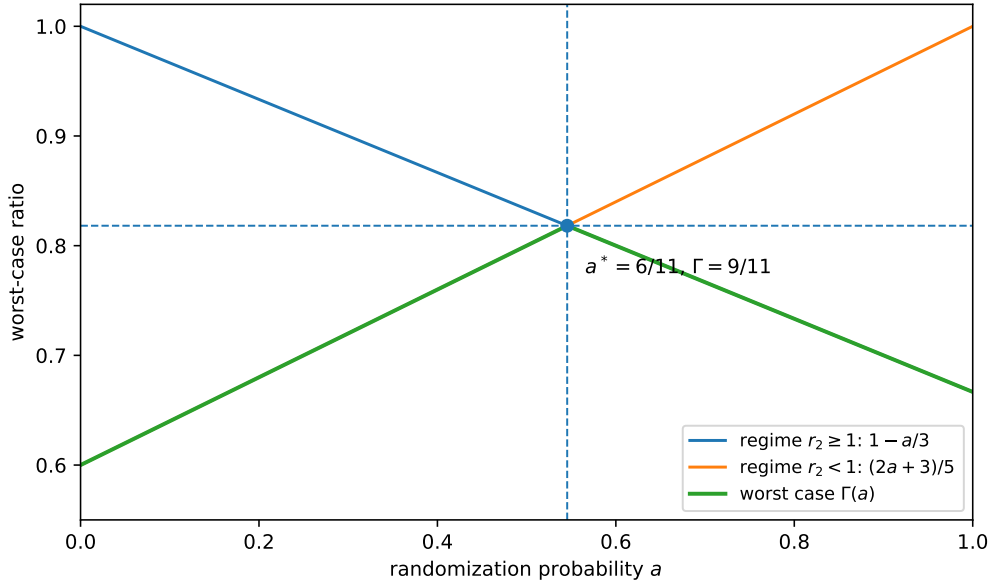


Figure 1: The two-period robust example of Proposition 1. The randomized rule balances the two adversarial regimes at $a^* = 6/11$, where the worst-case ratio is $9/11$.

2.6 Exact randomized minimax formulation

For general n , one should not expect a closed-form minimax randomized rule of the same simplicity as Bruss' threshold. Nevertheless, the exact robust randomized problem has a clean finite-dimensional/semi-infinite formulation. Let

$$\mathcal{U} = \prod_{i=1}^n \left[\frac{\bar{r}_i}{\rho}, \rho \bar{r}_i \right]$$

be the uncertainty set for the true odds. For a true instance $r = (r_1, \dots, r_n)$, define the value of the deterministic threshold s by

$$V_s(r) = \left(\prod_{j=s}^n \frac{1}{1+r_j} \right) \left(\sum_{j=s}^n r_j \right).$$

By Bruss' theorem,

$$V^*(r) = \max_{1 \leq s \leq n} V_s(r).$$

Let $\mathcal{D}_n$ be the finite set of deterministic stopping rules on the binary tree of histories. For $d \in \mathcal{D}_n$, let $W_d(r)$ be the probability that rule d stops on the last success under true odds r . A mixed randomized rule is a distribution $z \in \Delta(\mathcal{D}_n)$. Its value on instance r is

$$W_z(r) = \sum_{d \in \mathcal{D}_n} z_d W_d(r).$$

Thus the exact randomized minimax ratio is

$$\Gamma_n^* = \max_{z \in \Delta(\mathcal{D}_n)} \inf_{r \in \mathcal{U}} \frac{W_z(r)}{V^*(r)}.$$

Equivalently, Γ_n^* is the value of the semi-infinite linear program

$$\begin{aligned} & \text{maximize} && \gamma \\ & \text{over} && z \in \Delta(\mathcal{D}_n), \quad \gamma \geq 0 \\ & \text{subject to} && \sum_{d \in \mathcal{D}_n} z_d W_d(r) \geq \gamma V_s(r), \\ & && \forall r \in \mathcal{U}, \quad \forall s \in \{1, \dots, n\}. \end{aligned}$$

Indeed, the constraints for all s are equivalent to

$$W_z(r) \geq \gamma \max_s V_s(r) = \gamma V^*(r) \quad \forall r \in \mathcal{U}.$$

This formulation is exact but usually not closed-form. A practical approach is a cutting-plane algorithm: start with finitely many scenarios r , solve the resulting finite linear program, then search for a new scenario and threshold s violating the constraint most strongly. Adding this scenario and iterating gives increasingly tight randomized minimax rules. The two-period calculation above is the smallest case where this program can be solved by hand.

2.7 Naive target versus robust target

The naive idea is to ignore robustness and still target a predicted tail-odds sum of one:

$$\theta = 1.$$

Then $x = \rho$, and the robust discrete theorem gives

$$\frac{W_{\text{naive}}}{V^*} \geq \min \left\{ \frac{1}{\rho}, \left(1 + \frac{\rho - 1}{n - 1} \right)^{-(n-1)} \right\}.$$

For $\rho \geq 1$,

$$\left(1 + \frac{\rho - 1}{n - 1} \right)^{n-1} \geq 1 + (\rho - 1) = \rho,$$

so the second term is the smaller one. Therefore the finite- n naive guarantee is

$$\boxed{\beta_n(\rho) = \left(1 + \frac{\rho - 1}{n - 1} \right)^{-(n-1)}}. \tag{5}$$

By contrast, the optimized robust target uses

$$\theta_n^*(\rho) = \frac{x_n^*(\rho)}{\rho}.$$

For $\rho > 1$, one has $1 < x_n^*(\rho) < \rho$, hence

$$\theta_n^*(\rho) < 1.$$

Thus robustification starts slightly later than the naive predicted-Bruss rule. The reason is intuitive: if the true odds are larger than predicted, then starting at predicted mass one is too early and risks stopping on a success that is not the last.

2.8 Large- n limit

Let $n \rightarrow \infty$. For fixed x ,

$$\left(1 + \frac{x-1}{n-1}\right)^{n-1} \rightarrow e^{x-1}.$$

Hence (2) converges to

$$xe^{x-1} = \rho^2.$$

Equivalently,

$$xe^x = e\rho^2.$$

Therefore

$$\boxed{x_\infty^*(\rho) = W(e\rho^2), \quad \theta_\infty^*(\rho) = \frac{W(e\rho^2)}{\rho}, \quad \alpha_\infty(\rho) = \frac{W(e\rho^2)}{\rho^2}.} \quad (6)$$

The naive asymptotic guarantee is

$$\boxed{\beta_\infty(\rho) = e^{1-\rho}.} \quad (7)$$

For large ρ ,

$$\alpha_\infty(\rho) = \frac{W(e\rho^2)}{\rho^2} \sim \frac{2 \log \rho}{\rho^2},$$

whereas

$$\beta_\infty(\rho) = e^{1-\rho}$$

decays exponentially in ρ . Thus the robust target can be substantially better than blindly targeting one predicted odds unit.

Table 1: Discrete independent model. Optimized robust target versus naive target.

ρ	θ_{10}^*	α_{10}	β_{10}	θ_∞^*	α_∞	β_∞
1	1.000	1.000	1.000	1.000	1.000	1.000
1.25	0.989	0.792	0.781	0.988	0.790	0.779
1.5	0.967	0.645	0.615	0.962	0.642	0.607
2	0.911	0.455	0.387	0.900	0.450	0.368
3	0.805	0.268	0.164	0.782	0.261	0.135
5	0.656	0.131	0.037	0.618	0.124	0.018
10	0.465	0.047	0.002	0.418	0.042	0.000

3 Continuous Poisson model

We now move to the continuous model discussed by Bruss. Let successes arrive on $[0, T]$ according to an inhomogeneous Poisson process with true intensity $\varphi(t)$. Define the true remaining mass

$$M(t) = \int_t^T \varphi(u) du.$$

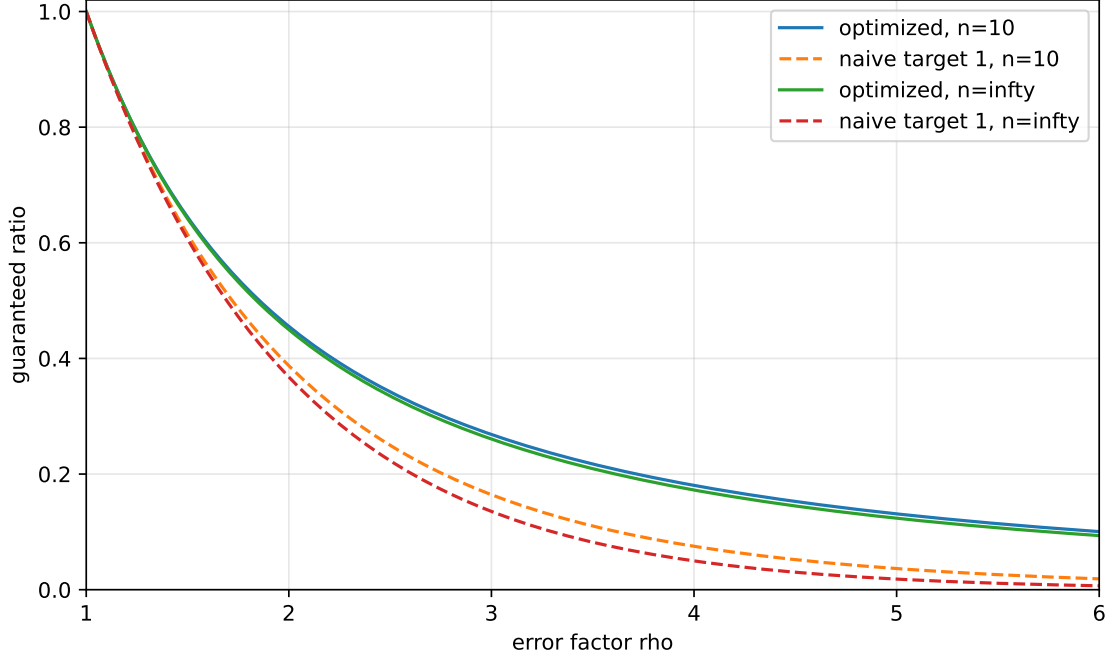


Figure 2: Discrete independent case: optimized robust target versus the naive target.

If we start accepting at time t and then stop at the first success after t , we win if and only if there is exactly one success after t . Since the number of future successes is Poisson with mean $M(t)$,

$$W(t) = \mathbb{P}(\text{Pois}(M(t)) = 1) = M(t)e^{-M(t)}.$$

The function $m \mapsto me^{-m}$ is maximized at $m = 1$, with value $1/e$. Thus, when the intensity is known, the optimal rule is

$$\boxed{\int_{\tau}^T \varphi(u) du = 1, \quad \text{then stop at the first success after } \tau.} \quad (8)$$

3.1 Robust intensity prediction

Suppose now that only a predicted intensity $\bar{\varphi}(t)$ is known, and that

$$\boxed{\frac{1}{\rho}\bar{\varphi}(t) \leq \varphi(t) \leq \rho\bar{\varphi}(t)} \quad \text{for every } t \in [0, T]. \quad (B)$$

Let

$$\bar{M}(t) = \int_t^T \bar{\varphi}(u) du.$$

A target θ means choosing τ_{θ} such that

$$\bar{M}(\tau_{\theta}) = \theta,$$

and then stopping at the first success after τ_{θ} . By (B), the true remaining mass satisfies

$$M(\tau_{\theta}) \in \left[\frac{\theta}{\rho}, \rho\theta \right].$$

Since the omniscient optimum has value $1/e$, the guaranteed ratio of target θ is

$$e \min_{m \in [\theta/\rho, \rho\theta]} me^{-m}.$$

Theorem 3 (Optimal robust Poisson target). *For $\rho > 1$, the robustly optimal target is*

$$\theta_P^*(\rho) = \frac{2\rho \log \rho}{\rho^2 - 1}. \quad (9)$$

The continuous extension gives $\theta_P^(1) = 1$. The resulting worst-case ratio against the omniscient optimum $1/e$ is*

$$\alpha_P(\rho) = \frac{2 \log \rho}{\rho^2 - 1} \exp\left(1 - \frac{2 \log \rho}{\rho^2 - 1}\right). \quad (10)$$

Proof. Let

$$f(m) = me^{-m}.$$

Then f is strictly increasing on $(0, 1)$, strictly decreasing on $(1, \infty)$, and maximized at $m = 1$.

Set

$$a = \frac{\theta}{\rho}, \quad b = \rho\theta = \rho^2 a.$$

For a fixed a , the uncertainty interval for the true mass is $[a, b]$. The robust value before multiplying by e is

$$\min_{m \in [a, b]} f(m).$$

We consider three possible regimes.

Regime 1: $b \leq 1$. Then the whole interval lies to the left of the maximizer. Since f is increasing on $(0, 1)$, the minimum is $f(a)$. Increasing a increases $f(a)$ while preserving $b \leq 1$ until $b = 1$. Thus the best point in this regime is $b = 1$, i.e. $a = 1/\rho^2$.

Regime 2: $a \geq 1$. Then the whole interval lies to the right of the maximizer. Since f is decreasing on $(1, \infty)$, the minimum is $f(b)$. Decreasing a decreases $b = \rho^2 a$, hence increases $f(b)$, until $a = 1$. Thus the best point in this regime is $a = 1$.

Regime 3: $a < 1 < b$. Then the minimum is

$$\min\{f(a), f(b)\}.$$

As a increases inside this regime, $f(a)$ increases and $f(b) = f(\rho^2 a)$ decreases. Hence the max-min point is where the two endpoint values are equal:

$$f(a) = f(\rho^2 a).$$

This is

$$ae^{-a} = \rho^2 ae^{-\rho^2 a}.$$

Since $a > 0$, cancellation gives

$$1 = \rho^2 e^{-(\rho^2 - 1)a}.$$

Taking logarithms yields

$$a^* = \frac{2 \log \rho}{\rho^2 - 1}.$$

Therefore

$$\theta_P^* = \rho a^* = \frac{2\rho \log \rho}{\rho^2 - 1}.$$

It remains only to check that this point indeed lies in Regime 3. We need

$$\frac{1}{\rho^2} < a^* < 1.$$

The upper bound is equivalent to

$$2 \log \rho < \rho^2 - 1,$$

which holds strictly for $\rho > 1$. The lower bound is equivalent to

$$\rho^2 \log(\rho^2) > \rho^2 - 1,$$

or, with $y = \rho^2 > 1$,

$$y \log y > y - 1,$$

which also holds for $y > 1$. Thus the equalizing point is feasible and it beats the boundary optima of the other regimes.

Finally, at the optimum the worst endpoint is

$$a^* = \frac{2 \log \rho}{\rho^2 - 1}.$$

The guaranteed probability is $a^* e^{-a^*}$, and the ratio against the omniscient value $1/e$ is

$$e a^* e^{-a^*} = a^* \exp(1 - a^*),$$

which is exactly (10). □

3.2 Naive Poisson target

The naive rule targets $\bar{M}(t) = 1$. Then the true remaining mass lies in

$$\left[\frac{1}{\rho}, \rho \right].$$

The guaranteed ratio is therefore

$$\boxed{\beta_P(\rho) = \min \left\{ \frac{1}{\rho} \exp \left(1 - \frac{1}{\rho} \right), \rho e^{1-\rho} \right\}.} \quad (11)$$

The optimized robust target satisfies

$$\theta_P^*(\rho) < 1 \quad (\rho > 1),$$

so, as in the discrete case, robustness means starting slightly later than the naive predicted-Bruss rule.

Table 2: Continuous Poisson model. Optimized robust target versus naive target.

ρ	θ_P^*	α_P	β_P
1	1.000	1.000	1.000
1.25	0.992	0.975	0.974
1.5	0.973	0.922	0.910
2	0.924	0.791	0.736
3	0.824	0.567	0.406
5	0.671	0.319	0.092
10	0.465	0.121	0.001

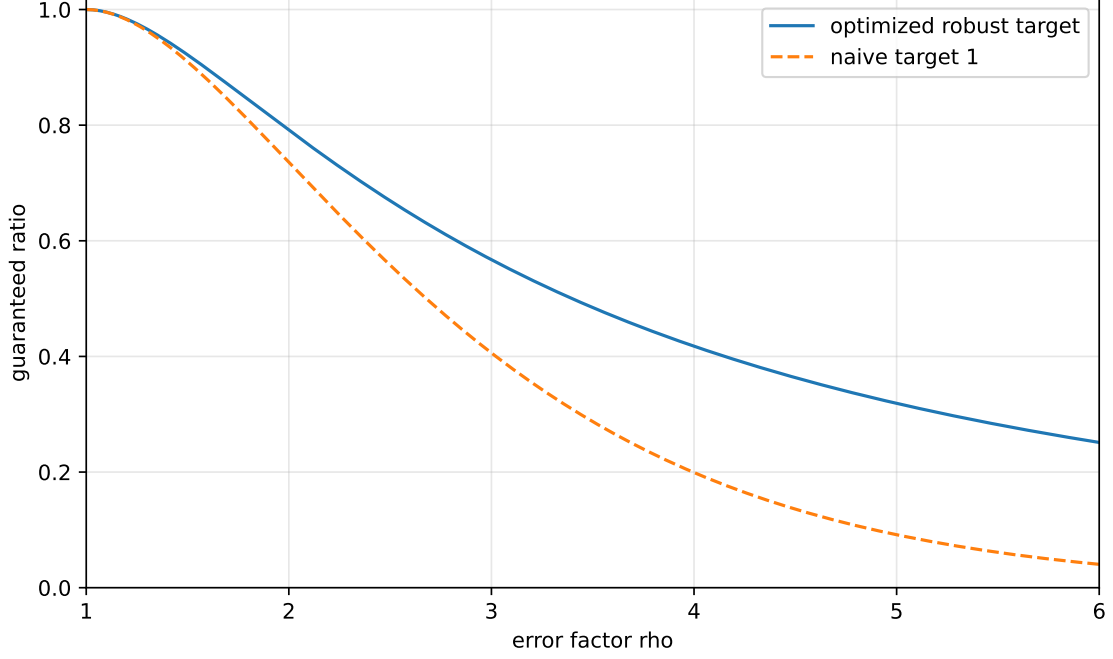


Figure 3: Continuous Poisson case: optimized robust target versus the naive target.

4 Bayesian random multiplicative errors on the odds

The preceding sections treated a worst-case multiplicative uncertainty set. We now consider a different, genuinely Bayesian, model. The prediction error is no longer bounded by a known factor and chosen adversarially. Instead, each local prediction error is drawn independently from a known law.

4.1 Model 2: independent random multiplicative errors

The model is the following. We observe predicted odds

$$\bar{r}_1, \dots, \bar{r}_n > 0.$$

For each time i , a hidden multiplicative error ρ_i is drawn independently from a known probability law $\mathcal{R}$ on $(0, \infty)$, and the true odd is

$$r_i = \frac{\bar{r}_i}{\rho_i}. \quad (12)$$

Equivalently, the prediction is

$$\bar{r}_i = r_i \rho_i.$$

Conditionally on ρ_i , the true success probability at time i is therefore

$$p_i(\rho_i) = \frac{r_i}{1 + r_i} = \frac{\bar{r}_i / \rho_i}{1 + \bar{r}_i / \rho_i} = \frac{\bar{r}_i}{\rho_i + \bar{r}_i}.$$

The indicators $I_1, \dots, I_n$ are generated independently conditionally on the hidden errors $\rho_1, \dots, \rho_n$.

The objective is the Bayesian one: maximize the conditional, ex ante probability of stopping on the last success given the observed predictions $\bar{r}_1, \dots, \bar{r}_n$ and the known error law $\mathcal{R}$.

Remark 2 (Local errors versus a global error). *This section uses independent local errors ρ_i . This assumption is crucial. If instead a single unknown factor ρ were shared by all times, then*

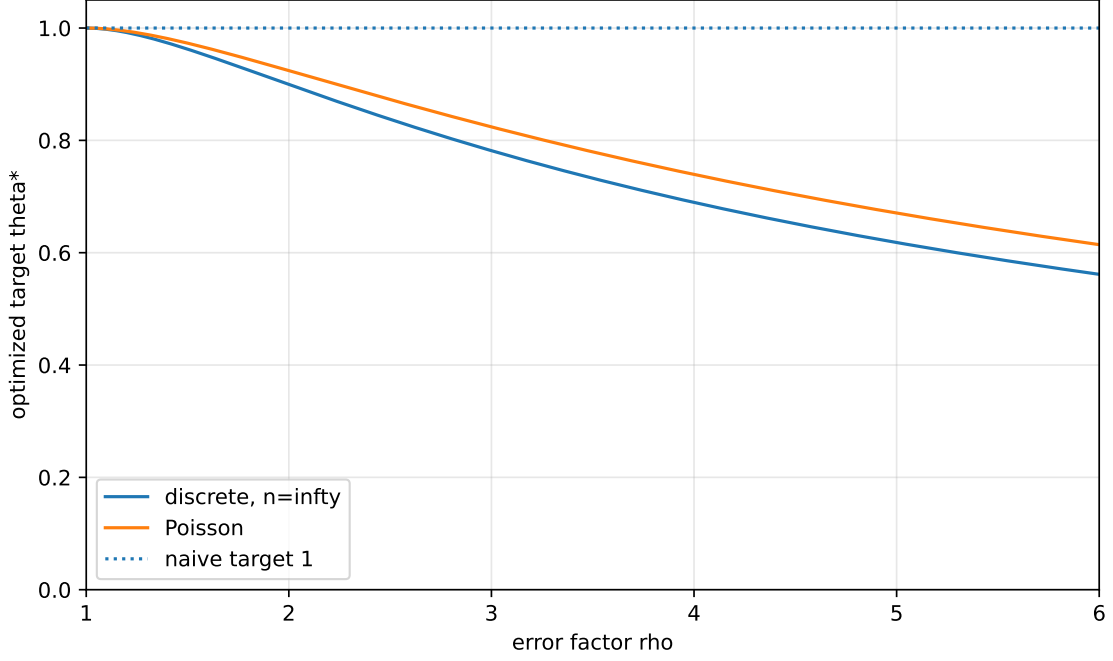


Figure 4: Optimized target levels. Both robust targets are below 1 as soon as $\rho > 1$: with uncertainty, one should start slightly later than the naive predicted-Bruss rule.

the observed successes and failures would update the posterior distribution of ρ , and the problem would become a dynamic program on that posterior. The simple reduction below would no longer apply.

4.2 The induced Bernoulli probabilities

Define the Bayesian marginal probability of success at time i by

$$\pi_i = \mathbb{E}_{\rho \sim \mathcal{R}} \left[\frac{\bar{r}_i}{\rho + \bar{r}_i} \right].$$

Then

$$1 - \pi_i = \mathbb{E}_{\rho \sim \mathcal{R}} \left[\frac{\rho}{\rho + \bar{r}_i} \right].$$

The corresponding Bayesian effective odd is

$$b_i := \frac{\pi_i}{1 - \pi_i} = \frac{\mathbb{E}_{\rho \sim \mathcal{R}} \left[\frac{\bar{r}_i}{\rho + \bar{r}_i} \right]}{\mathbb{E}_{\rho \sim \mathcal{R}} \left[\frac{\rho}{\rho + \bar{r}_i} \right]}. \quad (13)$$

Lemma 5 (Marginal independence after integrating the local errors). *Under Model 2, conditional on the predictions $\bar{r}_1, \dots, \bar{r}_n$, the indicators $I_1, \dots, I_n$ are independent Bernoulli random variables with success probabilities $\pi_1, \dots, \pi_n$.*

Proof. Fix a vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n) \in \{0, 1\}^n$. Conditionally on the hidden errors, the indicators are independent, hence

$$\mathbb{P}(I_1 = \varepsilon_1, \dots, I_n = \varepsilon_n \mid \rho_1, \dots, \rho_n) = \prod_{i=1}^n p_i(\rho_i)^{\varepsilon_i} (1 - p_i(\rho_i))^{1 - \varepsilon_i}.$$

Taking expectation over the independent hidden errors ρ_i factorizes the right-hand side:

$$\begin{aligned}\mathbb{P}(I_1 = \varepsilon_1, \dots, I_n = \varepsilon_n) &= \prod_{i=1}^n \mathbb{E} \left[p_i(\rho_i)^{\varepsilon_i} (1 - p_i(\rho_i))^{1-\varepsilon_i} \right] \\ &= \prod_{i=1}^n \pi_i^{\varepsilon_i} (1 - \pi_i)^{1-\varepsilon_i}.\end{aligned}$$

This is exactly the joint law of independent Bernoulli variables with parameters π_i . $\square$

Theorem 4 (Bayesian odds theorem under independent local errors). *Under Model 2, the Bayesian optimal rule is obtained by applying Bruss' odds algorithm to the effective odds b_i defined in (13). That is, define*

$$B_k = \sum_{j=k}^n b_j, \quad \tilde{Q}_k = \prod_{j=k}^n (1 - \pi_j),$$

with $B_{n+1} = 0$. In the non-degenerate case $B_1 > 1$, let

$$s_{\mathcal{R}} = \sup \{k : B_k \geq 1\}.$$

Then the Bayesian optimal rule ignores successes before $s_{\mathcal{R}}$ and stops at the first success from $s_{\mathcal{R}}$ onward. Its Bayesian value is

$$\boxed{V_{\mathcal{R}}^* = \tilde{Q}_{s_{\mathcal{R}}} B_{s_{\mathcal{R}}}.} \quad (14)$$

If $B_1 < 1$, the Bayesian optimal rule stops at the first success from the beginning.

Proof. By Lemma 5, after the hidden local errors have been integrated out, the Bayesian problem conditional on the predictions is an ordinary independent Bernoulli last-success problem with success probabilities π_i . Its odds are precisely

$$\frac{\pi_i}{1 - \pi_i} = b_i.$$

Bruss' odds theorem therefore applies verbatim to the sequence $\pi_1, \dots, \pi_n$. It says to sum the odds b_i from the end until reaching one, and then stop at the first success from that index onward. The value formula is the usual threshold value formula

$$\left(\prod_{j=s_{\mathcal{R}}}^n (1 - \pi_j) \right) \left(\sum_{j=s_{\mathcal{R}}}^n b_j \right) = \tilde{Q}_{s_{\mathcal{R}}} B_{s_{\mathcal{R}}}.$$

This proves the claim. $\square$

4.3 The calibration map

It is useful to view the theorem as a deterministic calibration of the predicted odds. For $x > 0$, define

$$M_{\mathcal{R}}(x) = \mathbb{E}_{\rho \sim \mathcal{R}} \left[\frac{1}{\rho + x} \right].$$

Then

$$\pi(x) = x M_{\mathcal{R}}(x), \quad 1 - \pi(x) = 1 - x M_{\mathcal{R}}(x),$$

and the effective odd associated with a predicted odd x is

$$\boxed{B_{\mathcal{R}}(x) = \frac{x M_{\mathcal{R}}(x)}{1 - x M_{\mathcal{R}}(x)}.} \quad (15)$$

Thus Model 2 is algorithmically simple:

$$\boxed{\bar{r}_i \mapsto B_{\mathcal{R}}(\bar{r}_i), \quad \text{then apply Bruss to these calibrated odds.}}$$

Lemma 6 (Basic properties of the calibration map). *For any law $\mathcal{R}$ on $(0, \infty)$, the map $x \mapsto B_{\mathcal{R}}(x)$ is nondecreasing and finite for every $x > 0$. Moreover, if $\mathbb{E}[1/\rho] < \infty$, then*

$$B_{\mathcal{R}}(x) \sim x \mathbb{E} \left[\frac{1}{\rho} \right] \quad (x \downarrow 0),$$

and if $\mathbb{E}[\rho] < \infty$, then

$$B_{\mathcal{R}}(x) \sim \frac{x}{\mathbb{E}[\rho]} \quad (x \rightarrow \infty).$$

Proof. The function $x \mapsto x/(\rho + x)$ is increasing for every fixed $\rho > 0$. Therefore

$$\pi(x) = \mathbb{E} \left[\frac{x}{\rho + x} \right]$$

is nondecreasing. Since $0 < x/(\rho + x) < 1$ for every $\rho > 0$, one has $0 < \pi(x) < 1$, hence

$$B_{\mathcal{R}}(x) = \frac{\pi(x)}{1 - \pi(x)}$$

is finite and nondecreasing.

If $\mathbb{E}[1/\rho] < \infty$, then

$$\frac{1}{x} \pi(x) = \mathbb{E} \left[\frac{1}{\rho + x} \right] \longrightarrow \mathbb{E} \left[\frac{1}{\rho} \right]$$

by monotone convergence. Since $\pi(x) \rightarrow 0$ as $x \downarrow 0$, this gives $B_{\mathcal{R}}(x) \sim \pi(x) \sim x \mathbb{E}[1/\rho]$.

If $\mathbb{E}[\rho] < \infty$, write

$$1 - \pi(x) = \mathbb{E} \left[\frac{\rho}{\rho + x} \right].$$

Then

$$x(1 - \pi(x)) = \mathbb{E} \left[\frac{x\rho}{\rho + x} \right] \longrightarrow \mathbb{E}[\rho]$$

by monotone convergence, because $x\rho/(\rho + x) \uparrow \rho$. Since $\pi(x) \rightarrow 1$, we obtain

$$B_{\mathcal{R}}(x) = \frac{\pi(x)}{1 - \pi(x)} \sim \frac{1}{1 - \pi(x)} \sim \frac{x}{\mathbb{E}[\rho]}.$$

□

4.4 Examples of error laws

The only quantity needed by the algorithm is $M_{\mathcal{R}}(x)$. For several simple laws it is explicit.

Deterministic multiplicative error. If $\rho \equiv c$, then

$$M_{\mathcal{R}}(x) = \frac{1}{c + x}, \quad B_{\mathcal{R}}(x) = \frac{x}{c}.$$

This recovers the obvious rescaling of all predicted odds.

Finite mixture. If $\rho = c_{\ell}$ with probability w_{ℓ} , where $\sum_{\ell} w_{\ell} = 1$, then

$$M_{\mathcal{R}}(x) = \sum_{\ell} \frac{w_{\ell}}{c_{\ell} + x},$$

and therefore

$$B_{\mathcal{R}}(x) = \frac{x \sum_{\ell} \frac{w_{\ell}}{c_{\ell} + x}}{1 - x \sum_{\ell} \frac{w_{\ell}}{c_{\ell} + x}}.$$

This is often the most practical case numerically: a continuous law can be approximated by a finite mixture, after which the algorithm is fully explicit.

Uniform law. If $\rho \sim \text{Unif}[a, b]$, with $0 < a < b$, then

$$M_{\mathcal{R}}(x) = \frac{1}{b-a} \log \left(\frac{b+x}{a+x} \right),$$

so

$$B_{\mathcal{R}}(x) = \frac{\frac{x}{b-a} \log \left(\frac{b+x}{a+x} \right)}{1 - \frac{x}{b-a} \log \left(\frac{b+x}{a+x} \right)}.$$

Gamma law. Let $\rho \sim \Gamma(a, \beta)$, where $a > 0$ is the shape and $\beta > 0$ is the rate:

$$f(\rho) = \frac{\beta^a}{\Gamma(a)} \rho^{a-1} e^{-\beta\rho}, \quad \rho > 0.$$

Then

$$\boxed{M_{\mathcal{R}}(x) = \beta^a x^{a-1} e^{\beta x} \Gamma(1-a, \beta x),} \quad (16)$$

where $\Gamma(\cdot, \cdot)$ denotes the upper incomplete gamma function. Hence

$$\boxed{B_{\mathcal{R}}(x) = \frac{\beta^a x^a e^{\beta x} \Gamma(1-a, \beta x)}{1 - \beta^a x^a e^{\beta x} \Gamma(1-a, \beta x)}. } \quad (17)$$

For the exponential case $a = 1$, this becomes

$$M_{\mathcal{R}}(x) = \beta e^{\beta x} E_1(\beta x),$$

where

$$E_1(y) = \int_y^\infty \frac{e^{-t}}{t} dt.$$

Shifted half-normal law. If

$$\rho = 1 + |Z|, \quad Z \sim \mathcal{N}(0, \sigma^2),$$

then $\rho \geq 1$, so the predictions always overestimate the true odds. In this case

$$\boxed{M_{\mathcal{R}}(x) = \sqrt{\frac{\pi}{2}} \frac{1}{\sigma} \exp \left(\frac{(x+1)^2}{2\sigma^2} \right) \text{erfc} \left(\frac{x+1}{\sqrt{2}\sigma} \right).} \quad (18)$$

Again $B_{\mathcal{R}}(x) = x M_{\mathcal{R}}(x) / (1 - x M_{\mathcal{R}}(x))$.

Log-normal law. A log-normal law is natural for multiplicative errors:

$$\rho = e^Z, \quad Z \sim \mathcal{N}(\mu, \sigma^2).$$

Then

$$M_{\mathcal{R}}(x) = \mathbb{E} \left[\frac{1}{x + e^Z} \right].$$

There is no particularly simple closed form in elementary special functions, but it is a one-dimensional integral and can be computed accurately by quadrature, for instance by Gauss–Hermite quadrature. Once the values $B_{\mathcal{R}}(\bar{r}_i)$ are computed, the stopping rule is still exactly the Bayesian Bruss rule of Theorem 4.

Table 3: Examples of the Bayesian calibration $B_{\mathcal{R}}(x)$. The exponential law has mean 1 but large variance and mass near 0, which inflates small effective odds. The shifted half-normal law satisfies $\rho \geq 1$, hence it mostly deflates the predicted odds.

$x = \bar{r}$	$\rho \equiv 1$	Unif[0.5, 1.5]	Exp(1)	$1 + N(0, 0.5^2) $
0.1	0.100	0.109	0.252	0.085
0.5	0.500	0.530	0.857	0.438
1.0	1.000	1.044	1.477	0.899
2.0	2.000	2.058	2.606	1.853
5.0	5.000	5.071	5.762	4.803

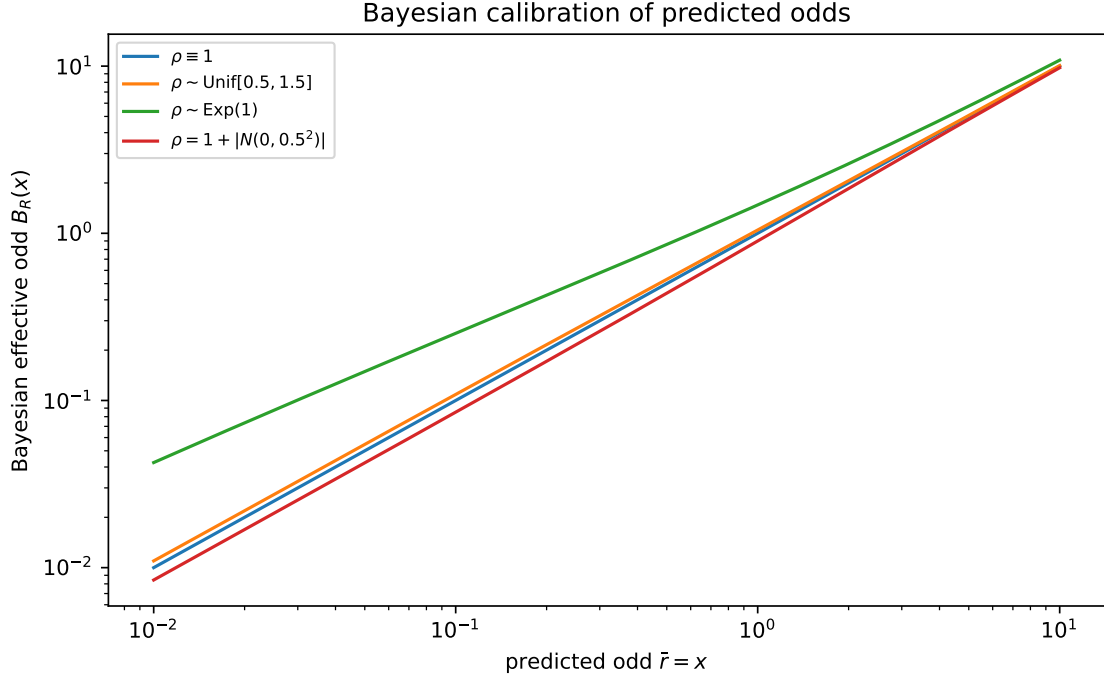


Figure 5: Bayesian calibration map $x \mapsto B_{\mathcal{R}}(x)$ for several error laws. After calibration, the optimal algorithm is simply Bruss' odds rule applied to the transformed odds.

4.5 What this model buys us

The worst-case robust model asks for a threshold that performs well for every realization of the multiplicative error up to a factor ρ . Model 2 asks a different question: if the local errors are random with known law, what rule maximizes the Bayesian success probability? The answer is especially neat: there is no need for a new stopping theorem. Random local multiplicative errors only change the Bernoulli probabilities. Once those induced probabilities are computed, Bruss' theorem applies without modification.

In short,

random independent local errors $\implies$ calibrate odds, then sum calibrated odds to one.

5 Interpretation

Known model. If the discrete odds r_i are known, the correct target is

$$\sum_{j=s}^n r_j \simeq 1.$$

If the Poisson intensity φ is known, the correct target is

$$\int_{\tau}^T \varphi(u) du = 1.$$

Predicted model with multiplicative error. If predictions can be wrong by a factor ρ , the naive target 1 is too optimistic. A robust rule targets a value below 1.

In the discrete independent model, the finite- n robust target is

$$\theta_n^*(\rho) = \frac{x_n^*(\rho)}{\rho}, \quad x_n^*(\rho) \left(1 + \frac{x_n^*(\rho) - 1}{n-1}\right)^{n-1} = \rho^2.$$

In the Poisson model, the robust target is

$$\theta_P^*(\rho) = \frac{2\rho \log \rho}{\rho^2 - 1}.$$

Why the Poisson guarantee is cleaner. The Poisson future is summarized by the scalar mass $M(t)$, and the success probability is exactly Me^{-M} . In the finite discrete case, an atom with a large odd can create overshoot effects, so the robust finite- n bound must also protect against the distribution of mass among finitely many indices. When all individual odds are small, the discrete model approaches the smoother Poisson behavior.

A Auxiliary inequalities used in the proofs

Lemma 7 (Product dominates one plus the sum). *For nonnegative $u_1, \dots, u_m$,*

$$\prod_{i=1}^m (1 + u_i) \geq 1 + \sum_{i=1}^m u_i.$$

Proof. Expand the product. All terms are nonnegative, and the expansion contains the constant term 1 and the linear terms $u_1 + \dots + u_m$. □

Lemma 8 (AM–GM product bound). *If $z_1, \dots, z_m \geq 0$ and $z_1 + \dots + z_m \leq S$, then*

$$\prod_{i=1}^m z_i \leq \left(\frac{S}{m}\right)^m.$$

Proof. This is the arithmetic–geometric mean inequality:

$$\left(\prod_{i=1}^m z_i\right)^{1/m} \leq \frac{1}{m} \sum_{i=1}^m z_i \leq \frac{S}{m}.$$

Raise both sides to the power m . □

References

- [1] F. Thomas Bruss. Sum the odds to one and stop. *The Annals of Probability*, 28(3):1384–1391, 2000.